

UCLA
Dept. of Electrical and Computer Engineering
ECE 214B: Advanced Topics in Speech Processing
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Project 2: Dementia Detection from Speech

Due: 6/2/2026

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I. Introduction

This project focuses on the automated detection of Probable Alzheimer's Disease (AD) from speech and language. Clinical screening tasks, such as picture description, elicit spontaneous connected speech that stresses lexical retrieval, coherence, and attention all of which degrade with AD. The goal is to establish acoustic and linguistic baselines to predict Control vs. Probable AD from speech segments and explore advanced techniques to improve detection performance.

II. Dataset

The DementiaBank Pitt Corpus [1] contains audio and transcripts from individuals with Probable Alzheimer's Disease and age-matched controls describing the "Cookie Theft" picture. Two modalities are available for each segment: 16 kHz mono WAV audio files and cleaned CHAT text transcripts. The provided dataset splits (train, dev, and test) are strictly speaker-disjoint, ensuring that no speaker appearing in the training set is present in the development or testing sets.

III. Project Codebase

The enclosed project package contains the necessary audio files, text transcripts, and split mappings (wav.scp, text, labels, and session metadata) required for the task. The accompanying Python notebook [2] contains a pipeline to establish segment-level predictions using frozen feature extraction and logistic regression, which are then aggregated to compute session-level probabilities.

IV. Objectives

The primary objective is to evaluate baseline acoustic and linguistic models for AD detection, and then extend upon them. Specifically, this involves an acoustic baseline using HuBERT-base [3], and a linguistic baseline using word 1- and 2-gram Term Frequency-Inverse Document Frequency (TF-IDF) [4], a statistic measuring a word's importance to a document. The baseline models predict at the segment level and aggregate to the session level by averaging per-segment probabilities. The project is structured into the following exploration tasks:

1. Experiment with other acoustic features, both embedding and non-embedding based (e.g., ECAPA-TDNN [5], WavLM [6], GeMAPS [7], HuBERT-large).
2. Explore other textual baselines. For example, replace TF-IDF with a pre-trained language

model like BERT [8] or RoBERTa [9].

3. Compare baseline performance across other demographic information (e.g., age, sex).
4. For the HuBERT acoustic baseline itself: explore representations from other layers (the baseline uses the last hidden layer).
5. Explore multimodal fusion (combining acoustic and textual features/predictions).
6. **Bonus:** The task is ultimately about clinical screening. Propose and implement an operating-point selection procedure that picks a threshold on dev maximizing F1 subject to recall >0.85 on the AD class, and report test precision/recall at that threshold.

V. Instructions

- A. Download the project package from [Box](#) [10] and upload it to your Google Drive account
- B. Make a copy of the [Colab notebook](#) [2]
- C. Open the notebook and run all the cells

VI. Baseline Results

The baselines utilize Logistic Regression with hyperparameter C swept on the dev session AUROC. The script should take around 15 minutes to run on Google Colab.

Method	Accuracy	F1	AUROC
Acoustic (HuBERT-base + LogReg)	68.7%	67.8%	0.772
Linguistic (TF-IDF 1-2gram + LogReg)	80.8%	80.5%	0.903

VII. Oral Presentations

There will be oral presentations by the different teams describing their work. Presentations should be planned by the team as a group.

VIII. Report and Code

The report (one per group) should include:

- Introduction (what is the problem/why is it important)
- Background (literature survey)
- Project Description (algorithm implementation, results, average run times, etc.)
- Summary and Discussion (also ideas for future work)
- References (cited throughout the report)

- Figures and flowcharts generally help clarify the text.

The report should be 4-pages long (excluding references) and have the same format as the INTERSPEECH conference. The presentations will be on **6/2** and the final reports may be turned in on **6/4**.

Useful References

1. DementiaBank Pitt Corpus:
Becker, J. T., Boiler, F., Lopez, O. L., Saxton, J., & McGonigle, K. L. (1994). "The natural history of Alzheimer's disease: description of study cohort and accuracy of diagnosis." Archives of Neurology, 51(6), 585-594.
2. Google Colab Notebook for feature extraction and classification:
<https://colab.research.google.com/drive/1D21QLAufomSZlv9PD6YleLmNBrAHrOEB?usp=sharing>
3. HuBERT model:
Hsu, Wei-Ning, et al. "HuBERT: Self-supervised speech representation learning by masked prediction of hidden units." IEEE/ACM Transactions on Audio, Speech, and Language Processing 29 (2021): 3451-3460.
4. TF-IDF:
Salton, Gerard, and Christopher Buckley. "Term-weighting approaches in automatic text retrieval." Information Processing & Management 24.5 (1988): 513-523.
5. ECAPA-TDNN
Desplanques, Brecht, Jenthe Thienpondt, and Kris Demuynck. "ECAPA-TDNN: Emphasized Channel Attention, Propagation and Aggregation in TDNN Based Speaker Verification." Proc. Interspeech (2020): 3830-3834.
6. WavLM
Chen, Sanyuan, et al. "WavLM: Large-scale self-supervised pre-training for full stack speech processing." IEEE Journal of Selected Topics in Signal Processing 16.6 (2022): 1505-1518.
7. GeMAPS
Eyben, Florian, et al. "The Geneva minimalistic acoustic parameter set (GeMAPS) for voice research and affective computing." IEEE Transactions on Affective Computing 7.2 (2015): 190-202.
8. BERT
Devlin, Jacob, et al. "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding." Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (2019): 4171-4186.
9. RoBERTa
Liu, Yinhan, et al. "RoBERTa: A Robustly Optimized BERT Pretraining Approach." arXiv preprint arXiv:1907.11692 (2019).

10. Package with all training and test files for the project:
<https://ucla.box.com/s/igex500j1a207hds7uksvdgwspp7c0vf>
11. ADReSS Challenge:
Luz, Saturnino, et al. "Alzheimer's dementia recognition through spontaneous speech: The ADReSS challenge." Proc. Interspeech 2020 (2020): 2172-2176.
12. Scikit-learn (TF-IDF & Logistic Regression):
Pedregosa, Fabian, et al. "Scikit-learn: Machine learning in Python." The Journal of Machine Learning Research 12 (2011): 2825-2830.